

Government of Maharashtra
Department of Agriculture
District Administration, Nandurbar

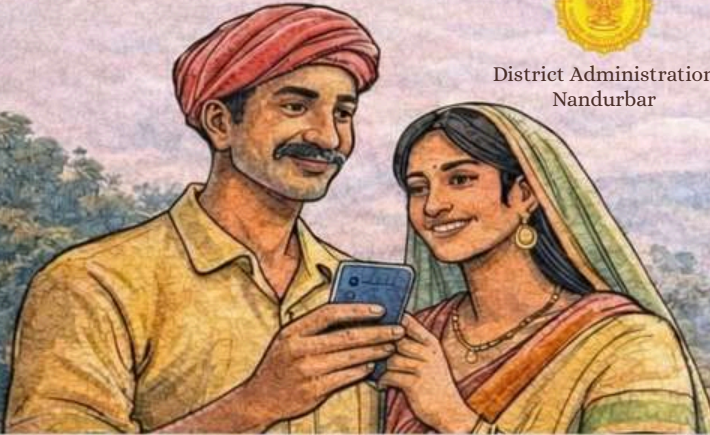
प्रकल्प अस्तित्व PROJECT ASTITVA

Democratising AI with Tribal Communities

A district-level replication handbook for
participatory language AI systems



District Administration
Nandurbar



Building Future-Ready Public Services

This handbook is an introductory guide to developing participatory AI systems for tribal and other underserved language communities.

Language is not just a medium of conversation. It's the key to accessing public information and trusting services.

Yet millions of people are excluded from critical services because they are not proficient in dominant languages.

By investing in multilingual AI, we can make government information and frontline support available in the languages citizens actually speak.

Today this handbook presents a democratic approach to digital communication, led by the district and built through broad institutional and community collaboration.

Project Astitva is a district-led mission to build participatory AI for local languages.

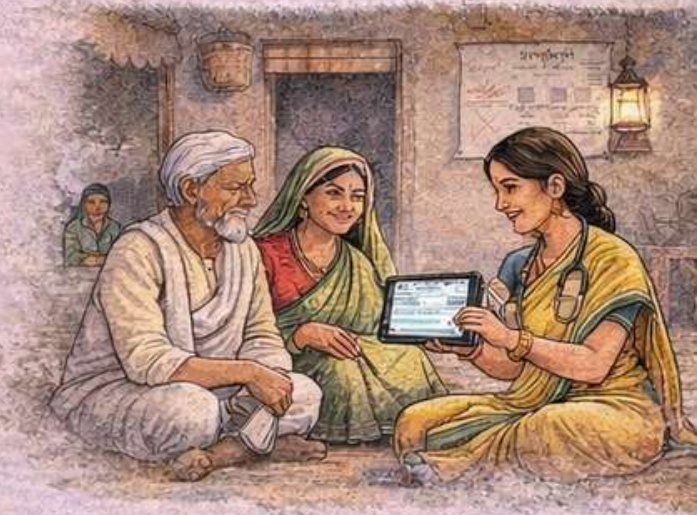


Empowering All Voices

Nandurbar District

A Model Site for Participatory Technology

Nandurbar is a predominantly tribal district where 69% of the population belongs to Scheduled Tribes (ST). The success and insights from Project Astitva in this vibrant, multilingual community set an example for inclusive development across India.



Linguistic & Tribal Diversity

Over 9 lakh tribal residents of Nandurbar speak over 18 languages and dialects, including Bhili, Pawari, Kokani, and Mavchi.

This journey begins in Nandurbar, a district where language, livelihood, and geography are deeply intertwined,



Participatory AI: A Core Principle

Participatory Artificial Intelligence (AI) is the foundation of inclusive digital governance for tribal and underserved language communities.

By involving communities directly in AI development, this approach ensures that both technology and policies reflect the needs and knowledge of the people they serve.

Inclusive Design



AI tools are co-created with local users, language experts, and frontline workers to ensure they are culturally relevant and understandable.

Collective Training



Community members actively contribute to creating and refining datasets, ensuring the data is accurate and reflective of their language.

Local Insight



Feedback loops incorporate local knowledge and dialects, allowing AI systems to continuously improve and remain relevant.

“Participatory AI is based on the philosophy that the best solutions are developed with the people, not for the people.”

A New Data Collection System

District + Karya Collaboration

The administration in Nandurbar collaborated with Karya, a social enterprise that specializes in creating ethical datasets for underserved communities. Together, we developed an innovative local language data collection system aimed at promoting voice-first AI, ensuring linguistic diversity and digital inclusion.

Planning and Coordination



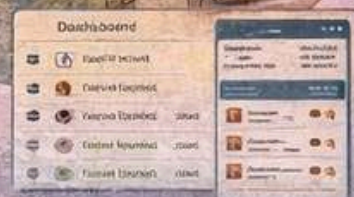
Guided by administration and local leaders.



Fieldwork and Data Collection



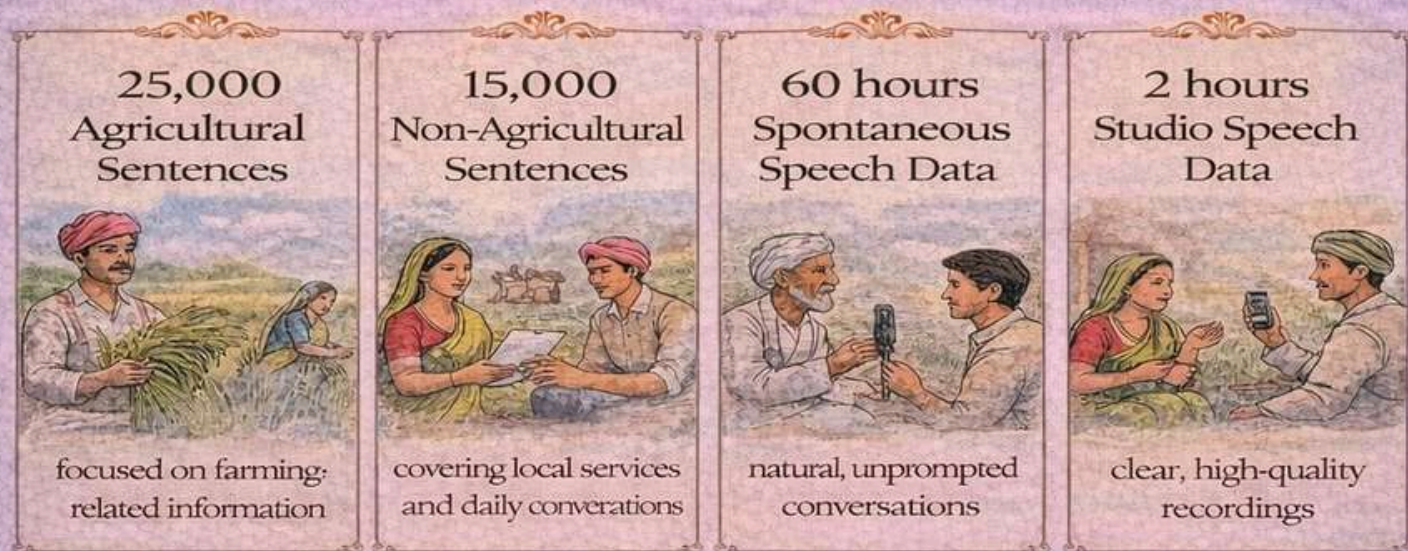
Dashboards, leaderboards, and engaging behaviour change communication "inspired "Tribaldaan", a community driven voice donation campaign for building voice datasets.



Through this joint effort, the district built capacity, Karya generated ethical employment, and the voice datasets helped build inclusive AI.”

What Data is Collected?

Through Project Astitva, Dehvali Bhili datasets were collected to integrate with MahaVISTAAR, the state's agricultural extension portal.



These 60,000+ voice samples in Dehvali Bhili will currently be integrated into MahaVISTAAR, Maharashtra's agricultural extension portal.



From Data to Deployment: MahaVISTAAR Integration

The collected Dehvali Bhili data followed a streamlined process to be integrated into MahaVISTAAR, Maharashtra's agricultural extension portal.

Recording → Processing → Integration → Results



Recording

Collection of Dehvali Bhili data using community-driven app



Processing

Speech-to-text conversion and quality checks ensuring accurate data



Integration

Accurate data integrated into MahaVISTAAR agricultural portal



Results

Enhanced access to farming services in local languages.

Participatory technology makes agricultural information accessible and relevant for Dehvali-speaking farmers.



MahaVISTAAR

State Agriculture Extension Platform

Early Outcomes and Field Response

The initial integration of Dehvali Bhili into the agriculture extension system marked the first practical deployment of language resources developed under Project Astitva.

Field demonstrations and pilot interactions showed that farmers engaged more confidently when information was delivered in a familiar language. Queries related to crop practices, weather conditions, and government advisories were received and understood with greater clarity.

Frontline workers and extension staff reported smoother communication during field visits and advisory sessions. The availability of local language support reduced the need for repeated explanations and informal translation.



Field Demonstration and Advisory Session

Key Learning

When communities participate in building digital systems, adoption improves and public trust deepens.

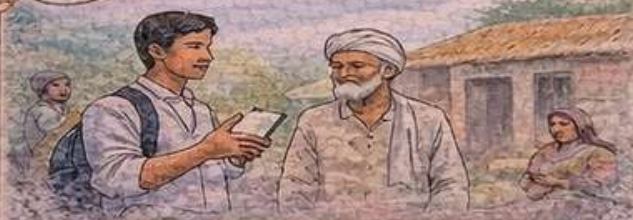
Early outcomes informed future scale-up and system refinement.

Replication Framework:

Steps to Empower More Districts

Scaling the Dehvali Bhili model to additional districts requires a flexible yet structured approach. Here's a step-by-step framework that can be adapted and replicated:

1 Assess Needs



Conduct a local assessment to identify linguistic and agricultural priorities.

2 Build Partnerships



Engage with local NGOs, cooperatives, and the Agriculture Department.

3 Develop Content & Tools



Adapt advisory content into local languages, using voice-based tools.

4 Deploy & Train



Launch the platform and conduct training for field staff and local resource persons.

Considerations and Lessons for Scale-Up

Expanding participatory language-based systems across Maharashtra presents both opportunities and challenges. Here are key considerations and lessons derived from the Dehvali Bhili experience:

1 Community Involvement

Involving local communities, especially women, ensures needs are met accurately and programs are well-received.



2 Customization

Each district has unique linguistic nuances, farming practices, and partner ecosystems. Solutions should be tailored accordingly.



3 Partnerships

Collaboration with NGOs, cooperatives, and government bodies facilitates information dissemination and sustainability.



These insights highlight the importance of adaptable frameworks, local leadership, and sustained evaluation for effective and scalable language interventions.



Insights from Implementation

The Following Section Documents Key Reflections and Practical Guidance drawn from field experience

Who Should Lead This Project?



The District Administration Should Lead

Participatory language AI initiatives require coordination across multiple departments, institutions, and community groups. The District Administration is uniquely positioned to bring these stakeholders and enable synergy between them.

In Maharashtra, delegated financial powers and innovative district level funds provide the flexibility to departments.

Since the District Collector heads all line departments, there is a unique opportunity to converge sectors—such as Education, IT, Agriculture, and Tribal Welfare—to ensure the entire administrative machinery works toward a unified developmental goal.

AI systems benefit them diversity of data across domains. Only a district level authority can enable pooling of such diverse data sources in a structured and lawful manner.

Why Leadership matters:

- Timely execution
- Cross department coordination
- Institutional ownership
- Sustained motivation across teams

Beyond Administration: Institutional Pride

Projects such as these are more than mere technical deployments; they represent a matter of district pride. When initiatives are driven by local leadership, they create a sense of confidence that ripples across all stakeholders.

Participatory AI requires participatory governance. District leadership make that possible

What Partnerships Are Required?

A Collaborative Ecosystem

District Administration

- Leadership
- Coordination
- Funding support

IIT Madras & EkStep

- Model development
- Technical validation

Karya

- Digital interface
- Dashboards and leaderboards
- Task management

Community Contributors

- Language data creation
- Validation
- Local ownership

State Agriculture Department & POCRA

- Service integration
- Deployment through extension systems

Bhashini

- Open source hosting
- National platform support

Sustained impact requires alignment between communities, institutions, and technology partners.

How Much Time and Resources Are Needed?

A Time Bound but Intensive Effort

The initial phase of Project Astitva was completed within one month.

- 25,000 agriculture sentences collected
- 15,000 non agriculture sentences collected
- 60 hours of spontaneous speech data recorded
- 6 hours of studio speech data recorded
- 2 hours of conversational speech data recorded
- Intensive coordination and sustained motivation
- were essential for meeting targets.



Diversity of Contributors Was Essential

- Agriculture Teachers
- Farmers
- Agriculture Health workers
- Revenue staff

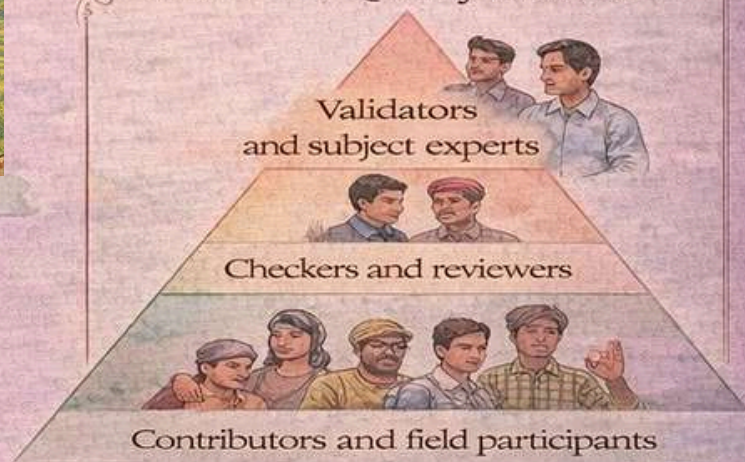
Multiple departments ^{staff} contributed to oversight and coordination.



Community Engagement Requires Time

Tribal contributors initially had concerns about spelling, scripts, and standardisation. Continuous communication was needed to build confidence and ensure clarity.

Three Tier Quality Structure



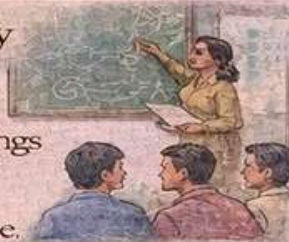
A similar structure was used for transcription efforts.

Successful language AI development requires coordinated effort across community, technical, and administrative layers.

Building Trust Before Building Technology

Step 1: Explain the Why

- Repeated field visits
- College and farmer meetings
- Open Q and A sessions
- Clearing fear around data use.



Understanding removed hesitation.

Step 2: Build Comfort and Ownership

- Icebreaking exercises
- Tribaldaan conclave with Bhashini
- Open discussion forums
- Public use of the term *Tribaldaan*



Participation became pride.

Step 3: Define Roles Clearly

- Only Dehvali Bhili contributors for this phase
- Elders as language anchors
- Youth as validators and digitizers
- Contributors registered on portal



Clarity prevented dilution.



Ethical compensation model ensured fair participation.

Building Trust



Central Model

Monitored sessions,
with coordinated collection

Decentralized Model

App based flexible
contributions from field



Participation was not assumed. It was designed, facilitated, compensated, and respected.

How Is Data Quality Ensured in Community Led Collection?

Ensuring Quality in Real Time



Robust Eligibility Criteria

Selection of the Right Contributors

- Proficiency in Dehwali Bhili
- Literacy and typing skills
- Local ownership in language standardisation.

Quality begins with selection.



Quality begins with selection.



Parallel Quality Tracks

Production and Validation Occurred Simultaneously

- Entries, checking and validation happened in real time.

3 parallel tasks tracked by digital dashboards



Speed and quality were balanced through quick identification, revalidation, and feedback

How Is Community Data Used Responsibly?

Ensuring Quality in Real Time



Clear Purpose of Open-Source Use

Transparency From the Start

- Data is used for open-source ASR, machine translation, and TTS.
- Bhashini platform ensures data openness for all.



Quality begins with selection.



Fair and Transparent Compensation

Ethical and Timely Payments

- Payments tied to verified contributions
- Direct transfers to registered accounts.



Consent and Transparency

Informed Participation as a Choice

- Participants registered through unique codes.
- Anonymized data provided privacy



Strong Safeguards Against Abuse

- Strict Monitoring and Accountability
- Irregularities addressed immediately



Respect for the community was paramount; openness, fairness and privacy were strictly maintained

How Is the System Governed and Sustained After Launch?

Institutionalization for Long Term Impact



Integration Into State Infrastructure

District work transitioned to the state.

- Institutional support from Mahavistaar
- Ongoing ownership by tribal department
- Transportability to other domains and districts.



National Open-Source Accessibility

Repository available on Bhashini

- Open access enables nationwide adaptability
- Ongoing updates and domain-specific expansions.
- Community language repository promotes sustained use.



Communities drive sustainability, state and national infrastructure support longevity.

What Are the Common Risks and How Can They Be Managed?

Community Mistrust



Fear and suspicion
around data collection



Mitigation: Transparency and repeated engagement



Poor Data Quality

Inaccurate, inconsistent data



Mitigation: Multi-layer validation



Motivation Fatigue

Drop in enthusiasm
over time



Mitigation: Dashboards, recognition, incentives



Administrative Overload

Burden on officers
managing initiatives



Mitigation: Timeline flexibility and leadership monitoring



Wrong Participation

Ineligible or uninterested
contributors



Mitigation: Clear eligibility criteria



Speed Over Quality

Limited Time can increase
speed but affect quality



Mitigation: Mid-course correction



Individual Dependency

Reliance on key
contributors



Mitigation: Open repositories



Absence of Baseline Resources

No existing database
to build upon



Mitigation: Centralized validated dataset

From One Language to Many Futures

Today

- ✓ Dehwali Bhili integrated into Mahavistaar
- ✓ Validated open language repository
- ✓ Community owned dataset



What Not to Rush

- ✓ Quality over speed
- ✓ Validation before scale
- ✓ Use case testing before expansion

Next Phase

- ✓ Three additional languages:
 - Mathwadi Bhili
 - Mavachi
 - Pawari



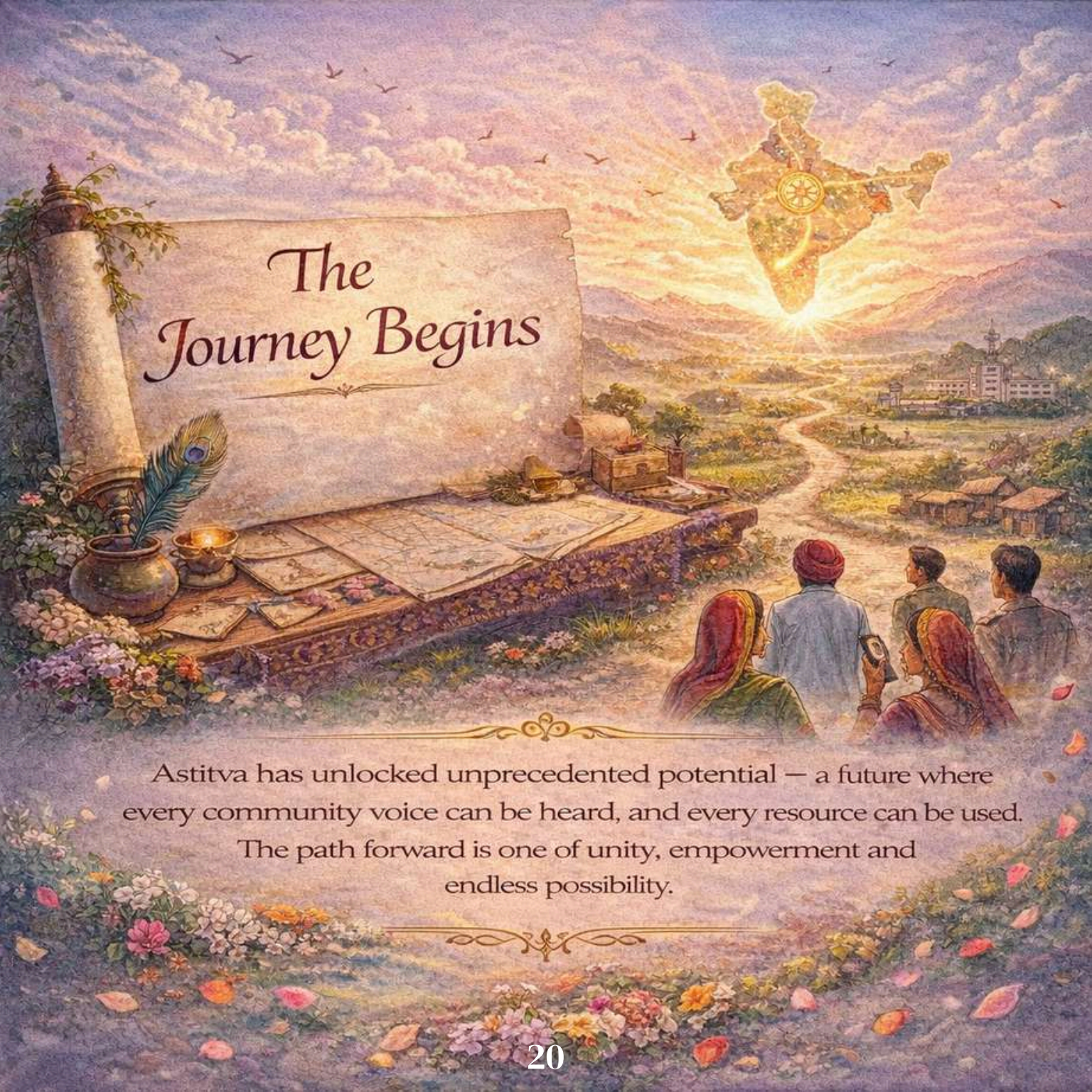
Long Term Vision

- ✓ A reusable public language infrastructure
- ✓ State-wide and national replication
- ✓ Every major service available in local language

Use Cases

- ✓ Asha / ANM / AWW Bot
 - Education
 - Administration

Once a language enters the digital ecosystem, possibilities multiply.



The Journey Begins

Astitva has unlocked unprecedented potential — a future where every community voice can be heard, and every resource can be used. The path forward is one of unity, empowerment and endless possibility.

